

QP Code : MCA23C101

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

III Semester - MCA C1 Internal Assessment January - 2026

Subject: Computer Applications

Title : Software Engineering

Time : 75 minutes

Max. Marks :30

PART A

I Answer the following: 2 × 10 = 20

1 What is software engineering? Explain software engineering code of ethics. CO1

OR

2 Explain the concept of risk management in project management. CO1

3 With a neat block diagram explain requirement elicitation and analysis process. CO2

OR

4 a Explain Data Flow Model with an example. CO2

b With a neat diagram explain the context model of a Hospital Management System.

PART B

II Answer the following: 2 × 5 = 10

5 With a neat diagram explain the waterfall model of software development process. CO1

OR

6 What is system dependability? Explain its four dimensions. CO1

7 What is requirement validation? Explain the various checks to be carried out during requirement validation. CO2

OR

8 Explain the state machine model of a microwave oven. CO2

QP Code : MCA23C201

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

III Semester MCA - C1 Internal Assessment : January - 2026

Subject : COMPUTER APPLICATIONS

Title : Machine Learning

Time : 75 minutes

Max. Marks :30

PART A

I Answer the following:

2 × 10 = 20

1. a. Explain the importance of machine learning.
- b. Write a note on issues in machine learning.

CO1 5

CO1 3

OR

2. a. With a neat diagram, explain ML framework.
- b. Explain the relation of ML with other fields.
3. a. Explain ID3 algorithm by solving a suitable example.

CO1 6

CO1 4

CO2 10

OR

4. a. Consider the given example,

CO2 10

Sr. No.	Maths	CS	Result
1	4	3	Fail
2	6	7	Pass
3	7	8	Pass
4	5	5	Fail
5	8	8	Pass

Classify the new data set

$x = (\text{Maths}=6, \text{CS}=8) (X_2, Y_2)$

using KNN algorithm and also discuss Distance Metrics used in KNN Algorithm.

PART B

II Answer the following:

2 × 5 = 10

5. Explain the examples of machine learning applications.

CO1 5

OR

6. Explain the process of designing a learning system.

CO1 5

PTO

7. Explain the types of decision trees.

CO2 5

OR

8. Consider the positively labelled data points $\{(3,1), (1,-1), (6,1), (6,-1)\}$ and negatively labelled data points $\{(1,0), (0,1), (0,-1), (-1,0)\}$. Find the separating hyperplane using Support Vector Machine.

CO2 5

QP Code : MCA23C208

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

III Semester MCA - C1 Internal Assessment : January - 2026

Subject : COMPUTER APPLICATIONS

Title : ETHICAL HACKING

Time : 75 minutes

Max. Marks :30

PART A

I Answer the following:

2 x 10 = 20

1. a. Explain the ethical hacking process and hacking methodology with suitable examples.
- b. Define hackers' behaviour and mindset.

CO1 6

CO1 4

OR

2. a. Describe information gathering techniques used in ethical hacking.
- b. Differentiate between active and passive sniffing.

CO1 6

CO1 4

3. a. Explain physical security vulnerabilities and their countermeasures in an organization.
- b. What is internal and external testing in penetration testing?

CO4 6

CO4 4

OR

4. a. Explain the preparation of ethical hacking and penetration testing reports and documents.
- b. State the importance of documentation in ethical hacking.

CO4 6

CO4 4

PART B

II Answer the following :

2 x 5 = 10

5. Explain social engineering attacks and their countermeasures.

CO1 5

OR

6. Describe IP spoofing and its impact on network security.
7. Explain wireless hacking and wireless foot printing techniques.

CO1 5

CO4 5

OR

8. Discuss DNS spoofing and network infrastructure vulnerabilities.

CO4 5

QP Code : MCA23C207

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

III Semester MCA - C1 Internal Assessment : January - 2026

Subject : COMPUTER APPLICATIONS

Title : Mobile Application Development

Time : 75 minutes

Max. Marks :30

PART A

I Answer the following:

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|----|----|---|--------------------|--|
| 1. | a. | Explain the Dart programming language and discuss its key features. | 2 × 10 = 20 | |
| | b. | Write a Dart program to demonstrate variables and data types | CO1 6 | |
| | | | CO1 4 | |

OR

- | | | | | |
|----|----|---|-------|--|
| 2. | a. | Explain data types in Dart and describe any three commonly used data type methods. | CO1 6 | |
| | b. | Explain standard input and output operations in Dart with an example. | CO1 4 | |
| 3. | a. | Explain the Flutter framework and discuss its benefits and scope in mobile application development. | CO4 6 | |
| | b. | List any four applications developed using Flutter with brief descriptions. | CO4 4 | |

OR

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|----|----|---|-------|--|
| 4. | a. | Explain the Flutter architecture and the role of widgets in Flutter applications. | CO4 6 | |
| | b. | Write short notes on Android Studio and VS Code as Flutter development tools. | CO4 4 | |

PART B

II Answer the following :

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|----|--|---|-------------------|--|
| 5. | | Explain control flow statements in Dart with suitable examples. | 2 × 5 = 10 | |
| | | | CO1 5 | |

OR

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|----|--|--|-------|--|
| 6. | | Explain Object-Oriented Programming (OOP) concepts in Dart | CO1 5 | |
| 7. | | Differentiate between Stateless Widget and Stateful Widget in Flutter. | CO4 5 | |

OR

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|----|--|---|-------|--|
| 8. | | Explain any five basic Flutter widgets such as Scaffold, Text, Image, AppBar, Column, or Row. | CO4 5 | |
|----|--|---|-------|--|

QP Code : MCA23C102

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

III Semester – MCA C1 Internal Assessment January - 2026

Subject : Computer Applications

Title: Python Programming

Time : 75 minutes

Max. Marks :30

PART A

I Answer the following: **2 × 10=20**

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|---|----|--|-----|---|
| 1 | a. | Explain Python data types. Classify them into mutable and immutable with examples. | CO1 | 6 |
| | b. | What will be the output of the following code?
a = 5
b = a
b = b + 2
print(a, b) | CO1 | 4 |

OR

- | | | | | |
|---|----|--|-----|---|
| 2 | a. | Explain input and output statements in Python with suitable examples. | CO1 | 6 |
| | b. | What is indentation? Explain its role in Python programs. | CO1 | 4 |
| 3 | a. | Explain lists in Python. Write a Python program to find the largest element in a list. | CO2 | 6 |
| | b. | Differentiate between list and tuple. | CO2 | 4 |

OR

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|---|----|--|-----|---|
| 4 | a. | Explain dictionaries in Python. Write a program to count the frequency of elements in a list using a dictionary. | CO2 | 6 |
| | b. | Write a short note on sets. | CO2 | 4 |

PART B

II Answer the following: **2×5=10**

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|---|--|--|-----|---|
| 5 | | Explain decision making statements in Python. Write a simple program using if-elif-else. | CO1 | 5 |
|---|--|--|-----|---|

OR

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|---|--|---|-----|---|
| 6 | | Write a Python program to print numbers from 1 to 10, but stop the loop when the number reaches 7 using the break statement.
Explain the role of the break statement and also explain its use in this program. | CO1 | 5 |
|---|--|---|-----|---|

- | | | | | |
|---|--|---|-----|---|
| 7 | | Define a function in Python. Explain default arguments with an example. | CO2 | 5 |
|---|--|---|-----|---|

OR

- | | | | | |
|---|--|--|-----|---|
| 8 | | Explain lambda functions. Write a program using map () or filter () to process a list. | CO2 | 5 |
|---|--|--|-----|---|

QP Code : MCA23C205

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE

III Semester MCA - C1 Internal Assessment : January - 2026

Subject : COMPUTER APPLICATIONS

Title : DIGITAL IMAGE PROCESSING

Time : 75 minutes

Max. Marks :30

PART A

I Answer the following: **2 × 10 = 20**

1. With a block diagram, explain the fundamental steps in image processing. CO1 10

OR

2. Write short notes on: CO1 10

i) Bit plane slicing

ii) Power law transformation.

3. Discuss how filtering of image is performed in frequency domain. CO2 10

OR

4. Discuss RGB, CMY and HSI color models in detail. CO2 10

PART B

II Answer the following: **2 × 5 = 10**

5. Write a short note on Ultrasound imaging. CO1 5

OR

6. Compute the lengths of the shortest 4-path, 8-path and m-path between the pixels p and q for $V = \{1, 2\}$. CO1 5

	3	1	2	1	— q
	2	2	0	2	
	1	2	1	1	
p —	1	0	1	2	

7. Discuss the working of an Average filter in image processing with suitable example. CO2 5

OR

8. Define Discrete Fourier Transform. Discuss various properties of 2D Discrete Fourier Transform. CO2 5
